AI304 PATTERN RECOGNITION PROJECT

Automated Face Recognition using DWT, PCA, and Machine Learning

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INTRODUCTION & PROBLEM STATEMENT

PATTERN RECOGNITION

Definition

Enabling computers to identify patterns, classify data, and make intelligent decisions based on observations.

Applications

Computer Vision, Medical Diagnosis, Biometrics, and Autonomous Systems.

`> Core field of Artificial Intelligence`

THE PROBLEM

Face Recognition Challenges

Accurately identifying individuals despite variations in expression, lighting, and head pose.

Project Motivation

Critical for modern security and human-computer interaction. Building a robust pipeline is essential.

`> Goal: Design an applied recognition system`

DATASET: AT&T DATABASE OF FACES

400
TOTAL IMAGES

40
DISTINCT SUBJECTS

Resolution: 64x64 grayscale pixels per image.

Variations: Captured variations in facial expression, lighting conditions, and head pose (up to 20 degrees).

Preprocessing: Images are standardized and normalized for consistency across the recognition pipeline.

Sample from Olivetti Dataset

PATTERN RECOGNITION PIPELINE

01

ACQUISITION

Loading the Olivetti Research Laboratory dataset.

SKLEARN DATA

02

FEATURE EXTRACTION

Decomposing images into frequency sub-bands.

2D DWT (HAAR)

03

DIM. REDUCTION

Projecting data onto principal components.

PCA (95% VAR)

04

CLASSIFICATION

Training multiple ML models for recognition.

SVM | KNN | RF

05

EVALUATION

Assessing performance with standard metrics.

F1 | ACCURACY

FEATURE EXTRACTION: DISCRETE WAVELET TRANSFORM

THE METHOD

Purpose

Decompose facial images into multi-resolution frequency components to isolate significant features.

Implementation

Applied a single-level 2D DWT using the Haar wavelet. `Wavelet: 'db1' (Haar)`

THE OUTPUT

Feature Selection

Extracted the **LL sub-band** (Approximation Coefficients) which contains the low-frequency structural data.

Key Benefit

Retains essential facial geometry while effectively filtering out high-frequency noise and minor variations.

`Output: Flattened 1D Vectors`

DIMENSIONALITY REDUCTION: PCA

THE CONCEPT

Principal Component Analysis

A statistical procedure that uses an orthogonal transformation to convert correlated variables into linearly uncorrelated principal components.

Feature Compression

Applied to DWT features to reduce the feature space while retaining the most significant information.

TECHNICAL IMPLEMENTATION

Variance Preservation

Configured to retain components explaining 95% of the total variance, ensuring minimal information loss.

> **95% Variance**
> Threshold for Component Selection

Eigenfaces

The resulting principal components represent the most prominent facial features across the dataset.

CLASSIFICATION MODELS

SVM (RBF)

A powerful classifier that finds the optimal hyperplane in high-dimensional space. Effective for complex decision boundaries.

CONFIGURATION

```
Kernel: RBF
C: 10
Gamma: 0.001
```

KNN (K=3)

A simple, instance-based learning algorithm that classifies based on the majority class of its nearest neighbors.

CONFIGURATION

```
Neighbors: 3
Metric: Euclidean
Weights: Uniform
```

RANDOM FOREST

An ensemble method using multiple decision trees to improve accuracy and control over-fitting.

CONFIGURATION

```
Estimators: 100
Criterion: Gini
Max Features: Sqrt
```

MODEL COMPARISON METRICS

MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE
SVM (RBF)	0.9500	0.9667	0.9500	0.9467
KNN (k=3)	0.8125	0.8083	0.8125	0.7842
Random Forest	0.9500	0.9708	0.9500	0.9483

VISUALIZATIONS: MODEL COMPARISON

KEY INSIGHTS

> PERFORMANCE PARITY

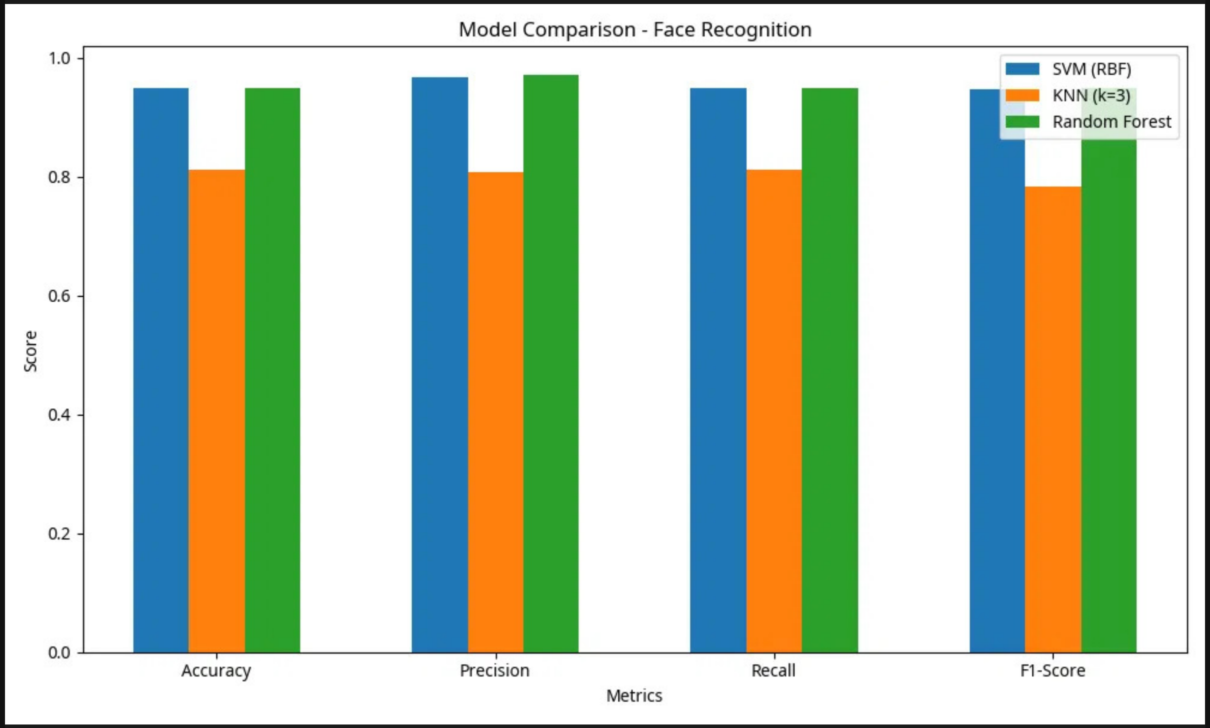
Both SVM and Random Forest achieved a peak accuracy of 95%, demonstrating high robustness in the feature space.

> KNN LIMITATIONS

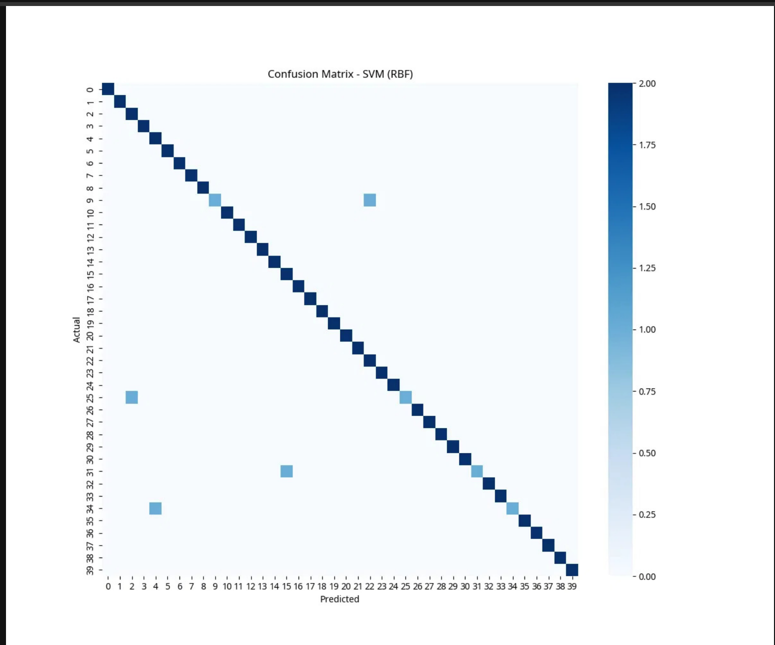
KNN lagged at 81.25%, likely due to the high-dimensional nature of facial data even after PCA reduction.

> PIPELINE SUCCESS

The combination of DWT and PCA provided a highly discriminative feature set for the top-tier classifiers.



VISUALIZATIONS: CONFUSION MATRIX



Model: Random Forest Classifier

ANALYSIS

Diagonal Density

The strong diagonal line confirms that the majority of test samples were correctly classified for all 40 subjects.

Error Distribution

Minimal off-diagonal points indicate very low misclassification rates between different individuals.

Robustness

The model demonstrates high reliability despite variations in facial expressions and lighting.

CONCLUSION & FUTURE IMPROVEMENTS

KEY FINDINGS

Pipeline Success

Successfully implemented a complete Pattern Recognition pipeline using DWT for feature extraction and PCA for dimensionality reduction.

Model Performance

SVM and Random Forest achieved a high accuracy of 95%, significantly outperforming KNN (81.25%).

`> All technical requirements fulfilled.`

FUTURE WORK

Advanced Architectures

Explore Deep Learning models (CNNs) and more complex wavelet families for improved feature representation.

Optimization & Scale

Implement automated hyperparameter tuning and evaluate performance on larger, more diverse datasets like LFW.

`> Target: Real-time robust recognition.`

Q&A

THANK YOU FOR YOUR ATTENTION!

Questions?